

LightGCN+ v4 — Reference Sheet

Description

Graph-based book recommender on the Book-Crossing dataset. Extends LightGCN with cold start handling so users with as few as 3 ratings still get recommendations — through demographic features (age, country) rather than graph propagation. Users with more than 10 ratings use full 4-layer graph propagation. Dataset includes 7,025 users and 4,576 cold users.

LightGCN+ v4 vs NeuMF Results

HR@5: 0.352 vs 0.268 (+31%)

HR@10: 0.463 vs 0.440 (+5%)

HR@20: 0.596 vs 0.590 (+1%)

NDCG@5: 0.263 vs 0.157 (+67%)

NDCG@10: 0.299 vs 0.213 (+40%)

NDCG@20: 0.333 vs 0.251 (+32%)

Cold Start Results

Cold users (≤ 10 train ratings) $n=4,576$: HR@10 = 0.403 | NDCG@10 = 0.265

Warm users (> 10 train ratings) $n=7,414$: HR@10 = 0.500 | NDCG@10 = 0.320

Cold users are included and handled — not filtered out.

Output Files

All saved in Downloads/results_v4/

lightgcn_v4_best.pt — saved model weights

results.json — all val + test metrics

error_analysis/report.json — final summary

error_analysis/diagnostics.csv — per-user data

error_analysis/01-09_*.png — all diagnostic plots